- Acorn Archimedes using ARM
- Acorn Phoeba 2100 using Intel StrongARM SA110

We end here with Intel for no other reason that to reiterate that even Intel at one point had an interest in something other than x86. These too a just a small sample of the many many options that came and went. These are just the more notable ones. Interesting times indeed. Let's see how we can make times interesting again now.

MESS and MAME

The answer is simpler than you might think. There is a project that has dedicated itself to being the one true emulation software for all such old and obsolete architectures and machines. This brilliant piece of technology calls itself MESS or Multi Emulator Super System.

MESS is actually a part of a project called MAME (Multiple Arcade machine Emulator). The MAME and MESS projects have been in development since 1997, and have over time accumulated the capability to emulate a large number of systems.

These projects have set their goal to preserving the history of computers and gaming by building emulators that document the working obsolete machines. The goal of this project isn't to make it easy or fast to run software for these systems, but just to serve as a historical record. For this they consider accuracy the most important criterion, above concerns of performance. As such they emulate even 3D rendering at the lowest levels in software when it is used, rather than using hardware acceleration which might only be an approximation.

Accuracy vs performance is actually a rather interesting discussion. Accuracy is obviously very important when it comes to a project that wants to keep a historical documentation of past systems. However for practical purposes performance is also a concern, especially when it comes to games. It might seem unimportant considering that many of those games ran on systems that were under 10MHz in speed, and modern systems are in the order of 2000 to 4000 MHz, even mobiles are generally over 1000 Mhz. However completely accurate emulation can be performance intensive. MAME goes to the extend of simulating signals flow through analogue sound circuits to get an accurate representation of sound. If this makes MAME / MAME less suitable for playing games, so be it, its job is no less important.

Currently MESS supports nearly a thousand (962) "unique systems with 2,063 total system variations" with more being developed and added. MAME